

# Week 2, Lecture 3 Handout: Advanced CSS Layouts & Cloud Deployment

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## ⚙️ Part 1: Advanced CSS Control (Positioning & Layering)

Before we publish our code to the world, we must master how to take elements completely out of the normal page flow and layer them creatively.

### 1. The `position` Property

By default, elements stack statically one after another. The `position` property alters this rule:

- **`position: relative;`** The element remains in the normal document flow, but you can nudge it away from its original spot using `top`, `bottom`, `left`, or `right` without affecting neighboring boxes. More importantly, it acts as a **structural anchor parent** for absolute children.
- **`position: absolute;`** The element is completely removed from the page flow. It floats freely and positions itself precisely relative to its nearest ancestor that has `position: relative;`. If no relative parent exists, it positions itself relative to the entire browser window (`<body>`).

### 2. The `z-index` Layering Engine

When you move elements around using absolute positioning, they will eventually overlap.

- **`z-index`** controls the 3D stacking order along the screen depth axis.
- An element with `z-index: 10;` will sit visually on top of an element with `z-index: 1;`.
- *Note: `z-index` only works on elements that have their position property explicitly set to `relative`, `absolute`, or `fixed`.*

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## 🔒 Part 2: Cloud Version Control (Git & GitHub)

Writing code locally is only half the battle. Professional developers must back up their work and publish it so users can access it online.

```
[ Working Directory ]  ---> ( git add ) ---> [ Staging Area ]
      (VS Code)                                (Ready to Save)
                                           |
                                           ( git commit )
                                           v
[ GitHub Cloud Server ] <--- ( git push ) <--- [ Local Repository ]
```

(Live Website)

(.git Database)

## 1. Core Definitions

- **Git:** A local tracking engine installed on your computer that monitors code changes line-by-line.
- **GitHub:** A cloud-based hosting platform that stores your tracked code repositories online.
- **Repository (Repo):** A digital project folder tracked by Git, containing all assets and history saves.

## 2. The International Bootcamp Deployment Commands

To push a local codebase into the cloud, you run this exact terminal pipeline:

1. `git init` – Turns your local folder into a tracked Git repository.
  2. `git add .` – Places all current changes into the Staging holding zone.
  3. `git commit -m "feat: design layout"` – Takes a permanent snapshot save of the staged changes.
  4. `git branch -M main` – Renames your default production line to the industry-standard name: main.
  5. `git remote add origin <URL>` – Creates a secure, direct link between your computer and your online GitHub folder.
  6. `git push -u origin main` – Uploads your local project snapshots into the cloud repo.
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## Homework Assignment: The International Capstone Deployment

**Objective:** Apply advanced layout elements to your profile project, track the workspace using local Git checkpoints, and publish it live on the internet via **GitHub Pages**.

### Task Instructions

#### 1. Refine Your Layout (CSS):

- Open your project folder inside **VS Code**.
- Set your `.form-card` container to `position: relative;`.
- Create a decorative badge element inside your HTML: `<span class="badge">Live</span>`.
- Target `.badge` in your CSS and apply `position: absolute; top: 20px; right: 20px; z-index: 5; background: #e74c3c; color: white; padding: 4px 8px; border-radius: 4px;`. This layout trick cleanly places the floating badge right in the top-corner of your container box.

#### 2. Initialize & Track (Git):

- Open your VS Code built-in Terminal.

- Initialize local tracking by running: `git init`
  - Stage and commit your layout modifications safely:

```
git add .  
git commit -m "feat: complete design layout with  
positioning"
```

### 3. Go Cloud (GitHub):

- Log into GitHub and create a brand new **Public** repository named `my-profile-page`. Do not check any initialization boxes (leave README unchecked).
- Copy the existing repository command lines displayed on your browser screen.
- Paste those commands straight into your VS Code terminal and press enter to push your code to the cloud.

### 4. Publish Live (GitHub Pages):

- On your GitHub repository webpage, navigate to the **Settings** tab at the top.
  - Scroll down the left-hand menu and click on **Pages**.
  - Under the "Build and deployment" section, change the Branch dropdown selection from *None* to **main**, leave the folder as `/ (root)`, and click **Save**.
  - Wait 60 seconds and refresh the browser page. GitHub will generate a live public link (e.g., `https://github.io`).
  - Open that live link in a browser tab to confirm your fully styled web profile is officially accessible across the globe!
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